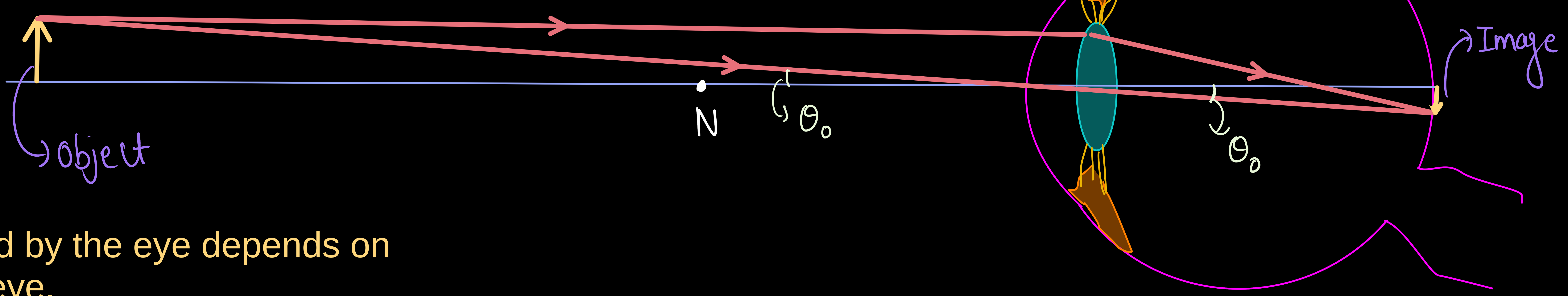
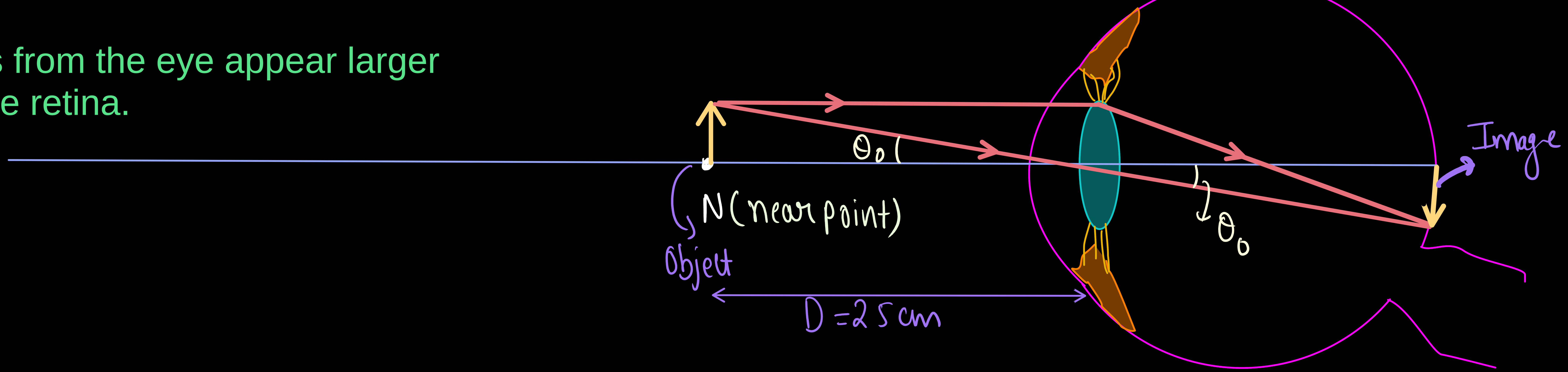
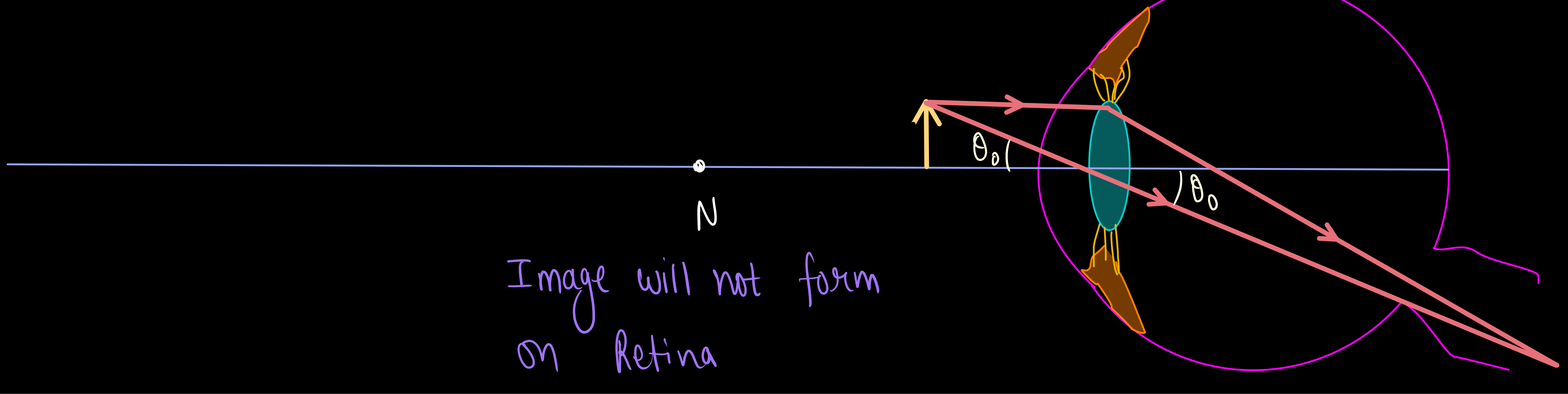
Simple Microscope



The apparent size of an object perceived by the eye depends on the angle the object subtends from the eye.

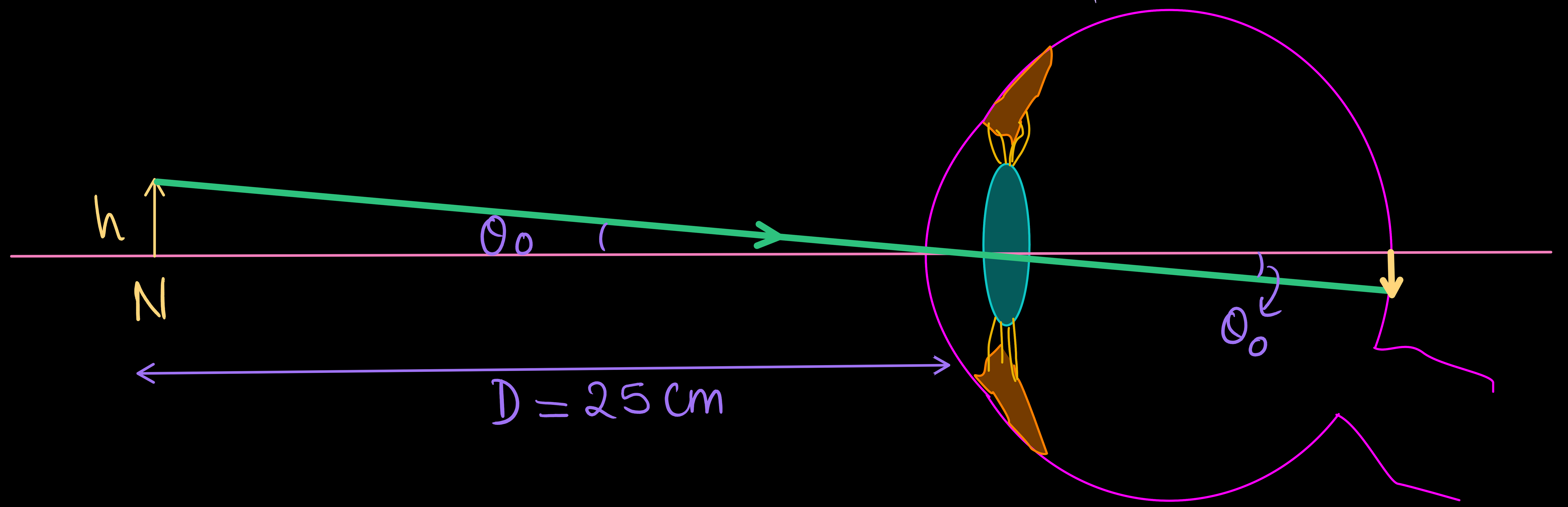
Thus, objects that subtend large angles from the eye appear larger because they form larger images on the retina.

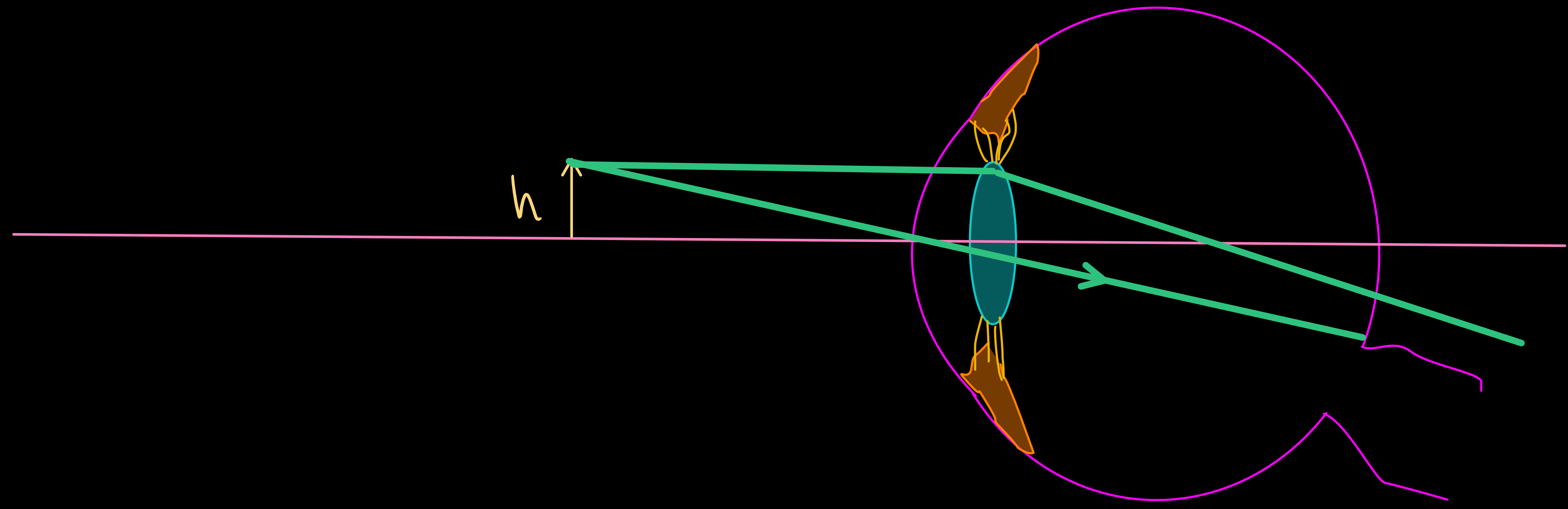




Simple Microscope

for naked eye –
The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.





Simple Microscope

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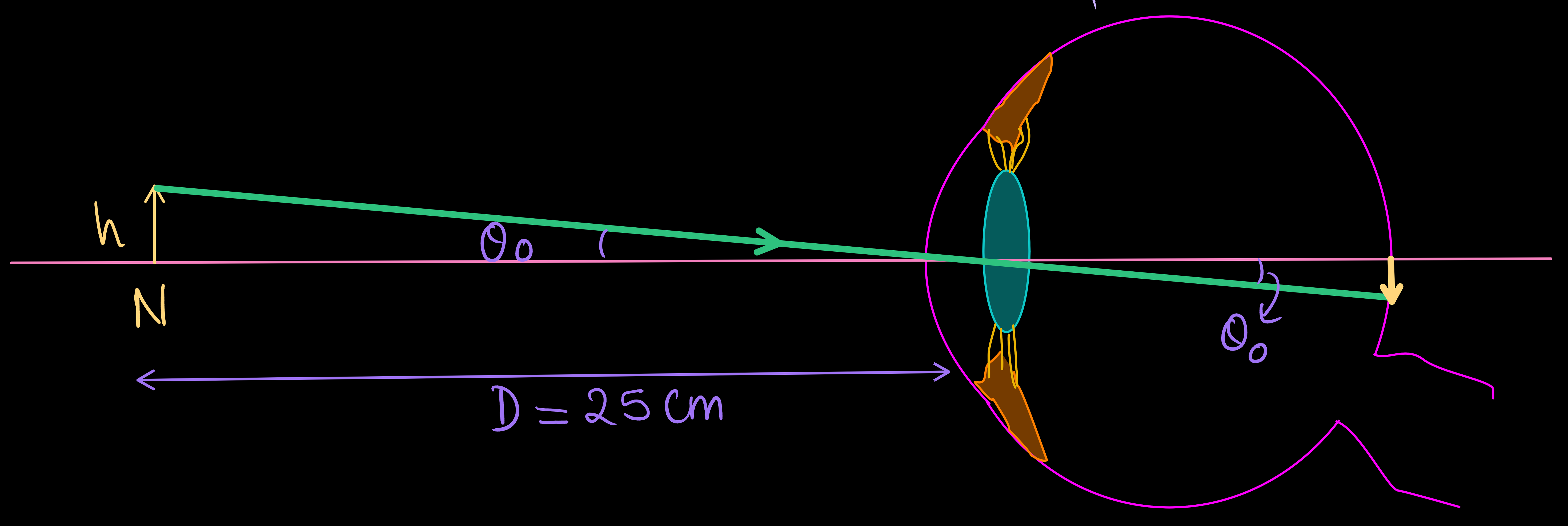
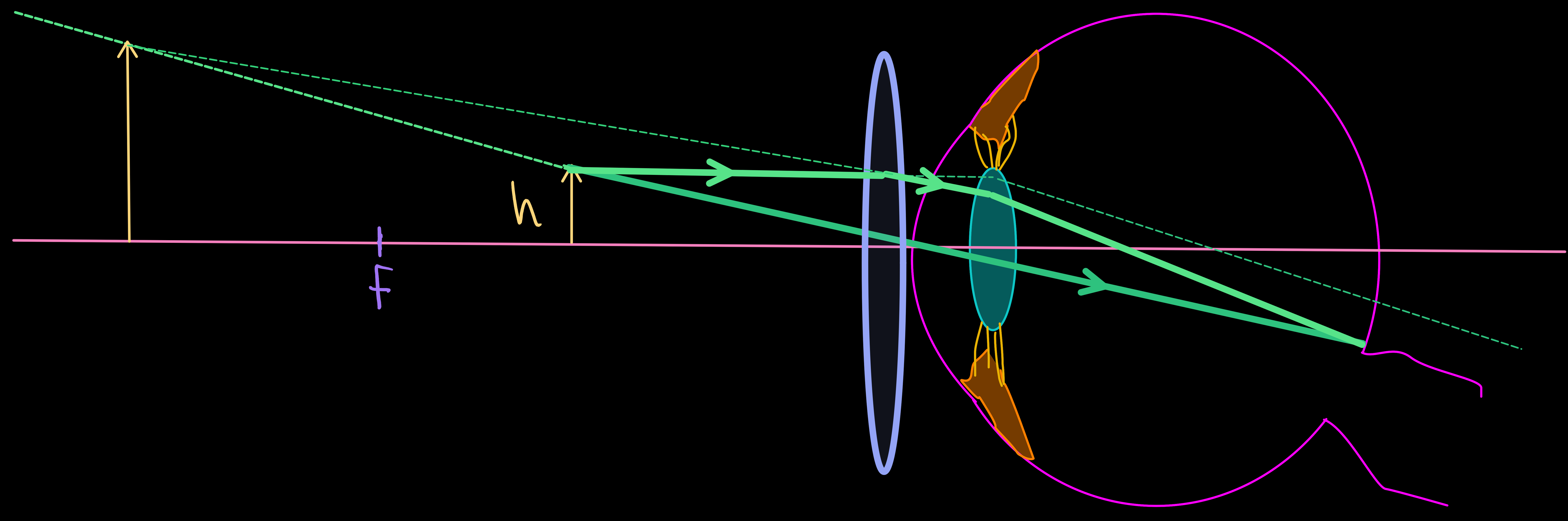


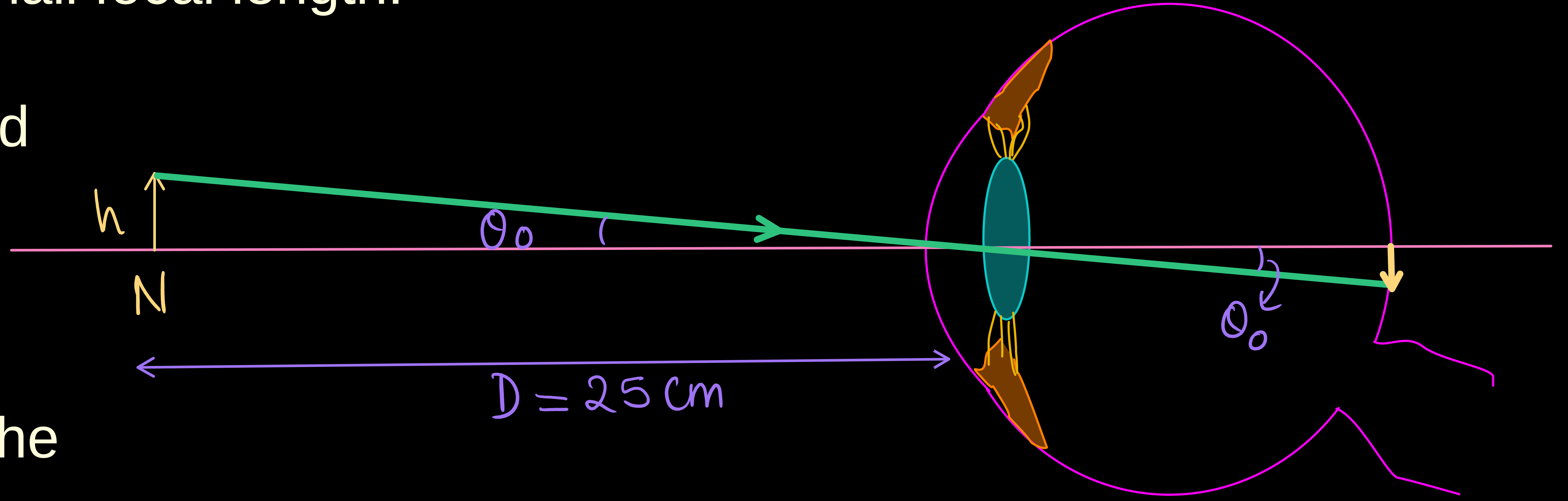


Simple Microscope

for naked eye –
The angular size (θ_o) of image formed at retina is maximum when object is at near point of the eye.

* A simple magnifier or microscope is a converging lens of small focal length.

* In order to use such a lens as a microscope, the lens is held near the object, one focal length away or less, and the eye is positioned close to the lens on the other side.



* The idea is to get an erect, magnified and virtual image of the object at a distance so that it can be viewed comfortably, i.e., at 25 cm or more.

Case I → When image is formed at Near point (D = 25 cm)

The max. angle subtended by the object at eye and can be clearly visible (without lens)

$$\theta_o = \frac{h}{D} \quad \text{—} \ (1)$$

Now, the angle subtended by image at eye.

$$\theta_i = \frac{h}{x} \quad \text{—} \ (2)$$

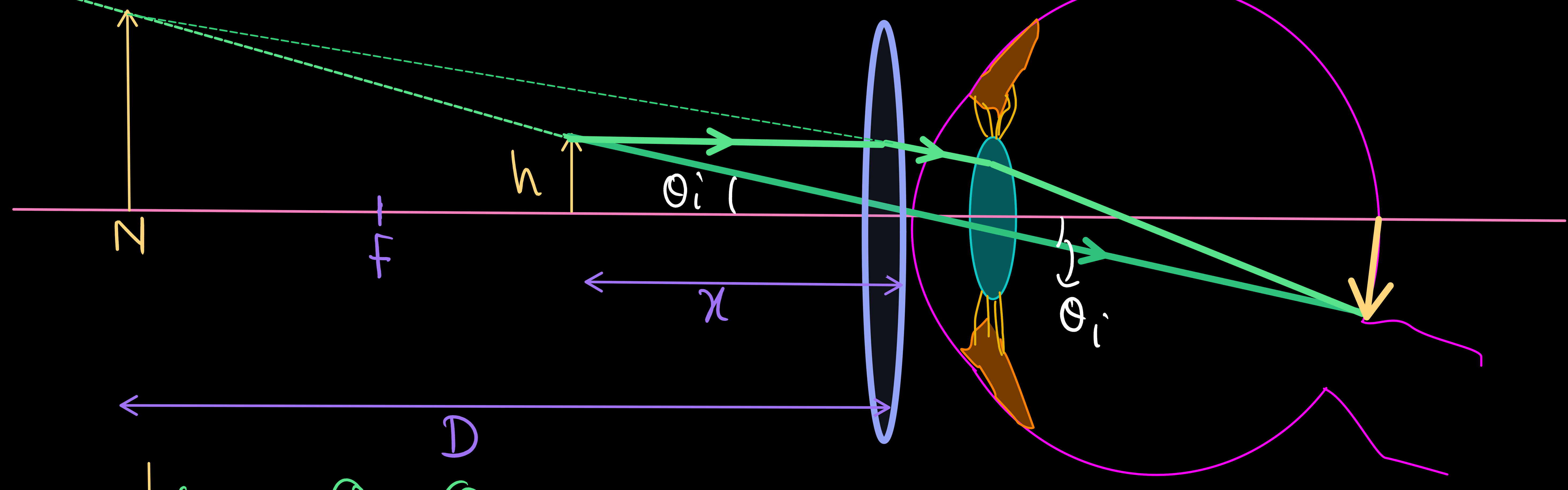
using lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$v = -D, \ u = -x$

$$\frac{1}{f} = \frac{1}{-D} - \frac{1}{-x}$$

$$\boxed{\frac{1}{f} + \frac{1}{D} = \frac{1}{x}} \quad \text{—} \ (3)$$

$\theta_i > \theta_o$

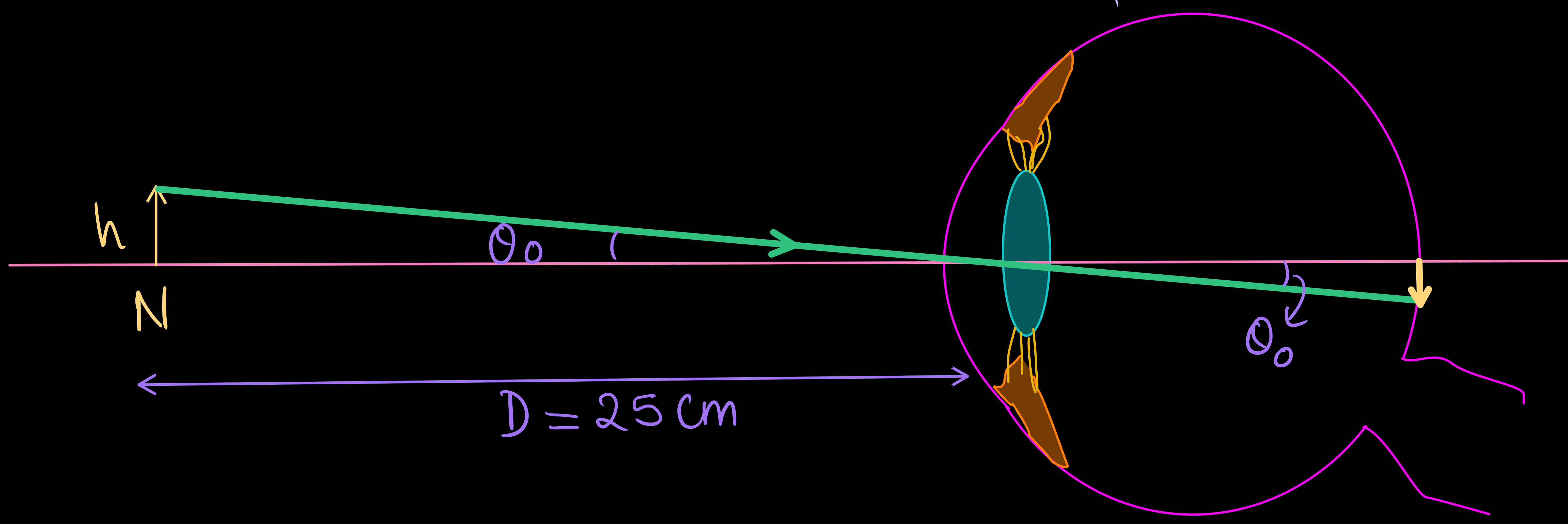


from eq (2) & (3)

$$\boxed{\theta_i = h\left(\frac{1}{f} + \frac{1}{D}\right)} \quad \text{—} \ (4)$$

Simple Microscope

for naked eye –
The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.



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$v = -D$, $u = -x$

$$\frac{1}{f} = \frac{1}{-D} - \frac{1}{-x}$$

$$\boxed{\frac{1}{f} + \frac{1}{D} = \frac{1}{x}} \quad \text{— (3)}$$

from eq (2) & (3)

$$\boxed{\theta_i = h\left(\frac{1}{f} + \frac{1}{D}\right)} \quad \text{— (4)}$$

Now, Angular Magnification

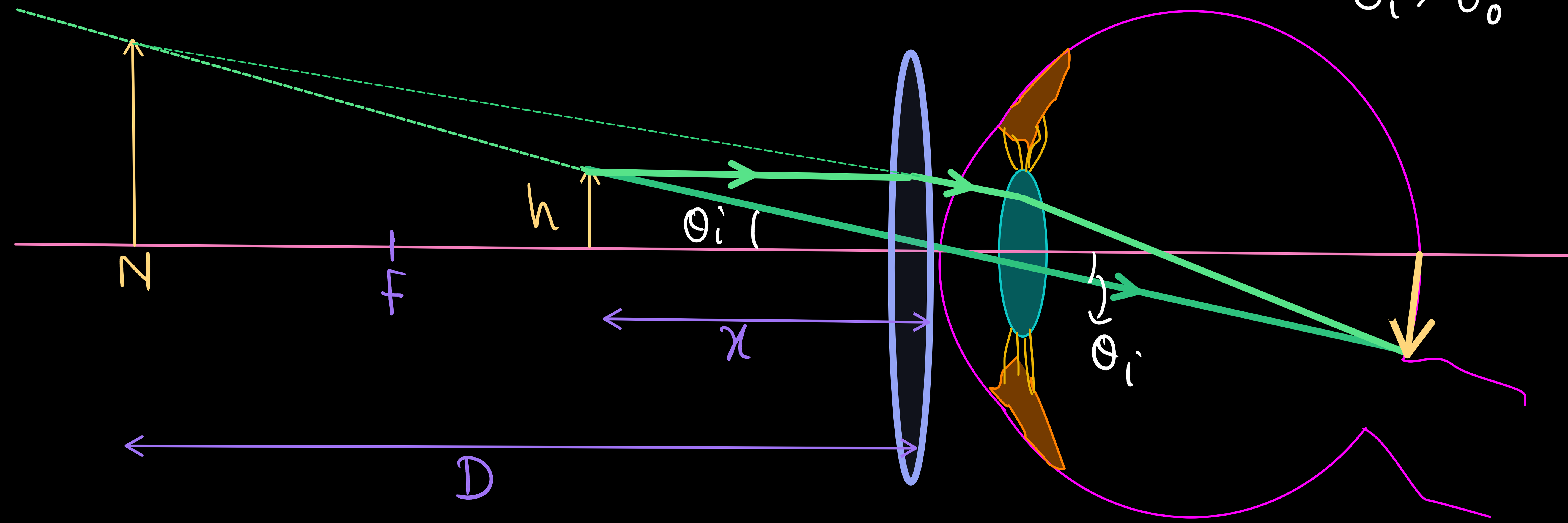
$$m = \frac{\theta_i}{\theta_0} = \frac{\cancel{h}\left(\frac{1}{f} + \frac{1}{D}\right)}{\frac{\cancel{h}}{D}}$$

$$m = D\left(\frac{1}{f} + \frac{1}{D}\right)$$

$$\boxed{m = 1 + \frac{D}{f}}$$ ★★

Magnification of simple microscope when image is formed at Nearpoint (Least Distance of distinct vision)

$\theta_i > \theta_0$

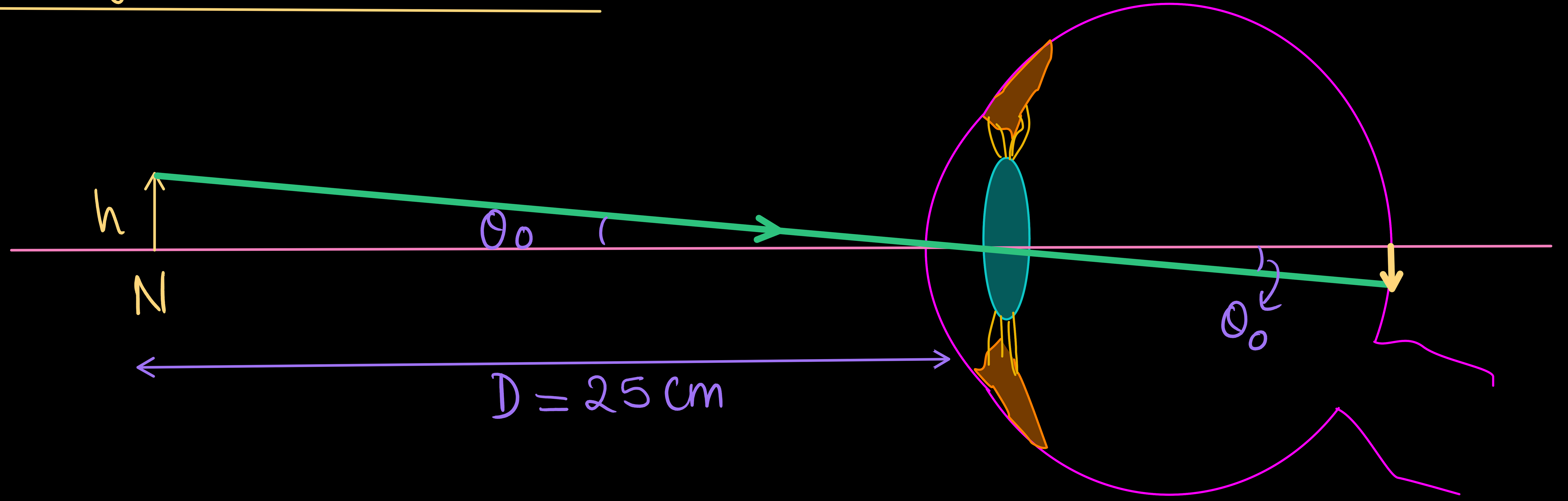


Simple Microscope

for naked eye –
The angular size (θ_0) of image formed at retina is maximum when object is at near point of the eye.

Case II – When image is formed at infinity (eye is relaxed, veiwing is more comfortable)

* The object is placed at focus of the convex lens to form the image at ∞.

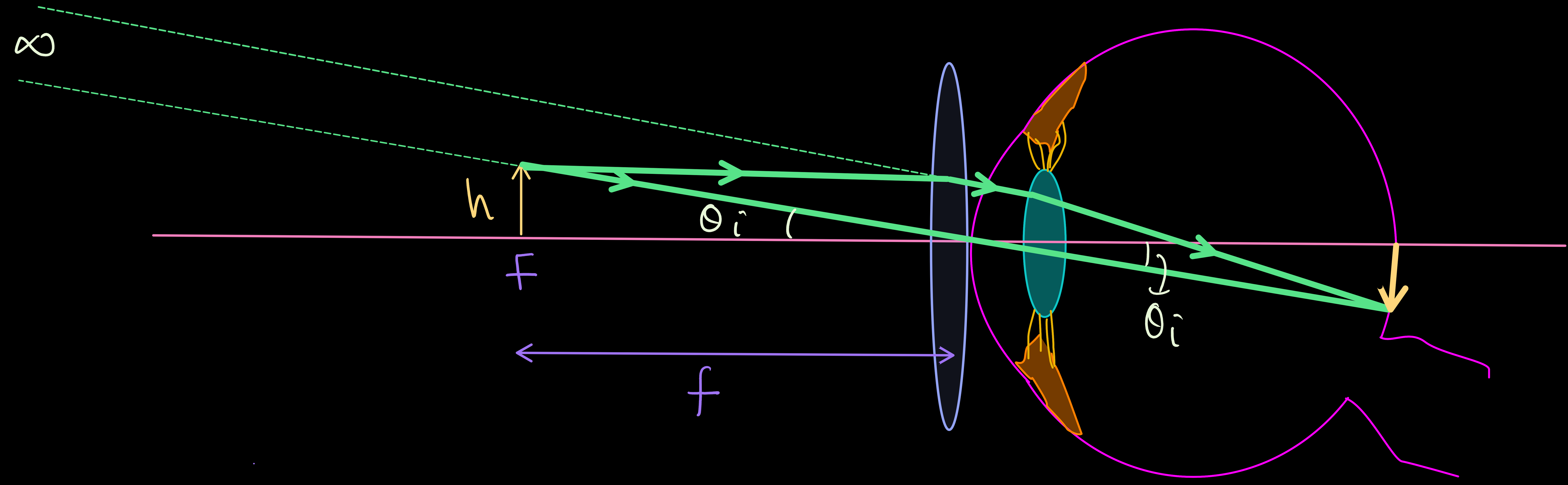


Angle subtended by object at eye (without lens)

$$\theta_0 = \frac{h}{D} \quad -①$$

Angle subtended by image at eye –

$$\theta_i = \frac{h}{f} \quad -②$$



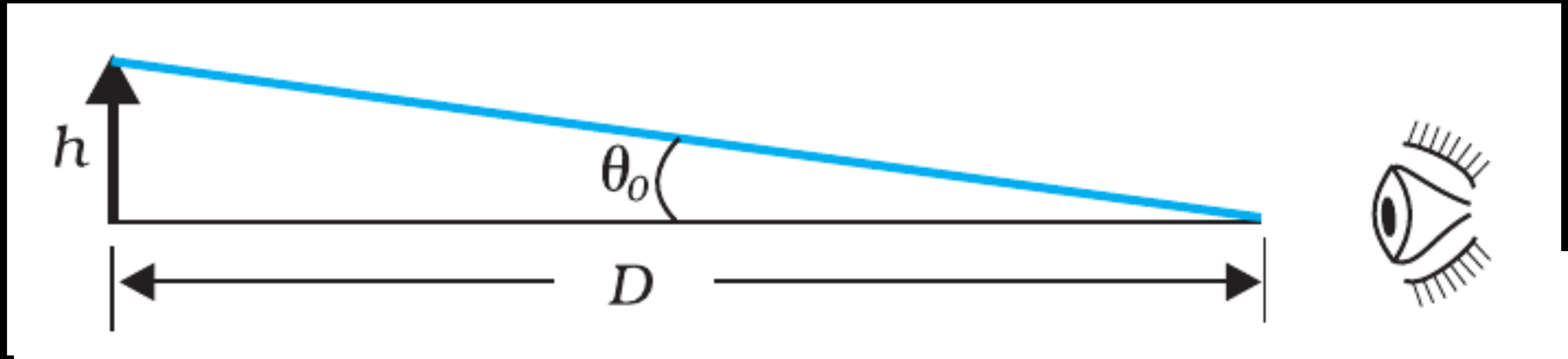
∴ Angular magnification produced by simple microscope in normal adjustement –

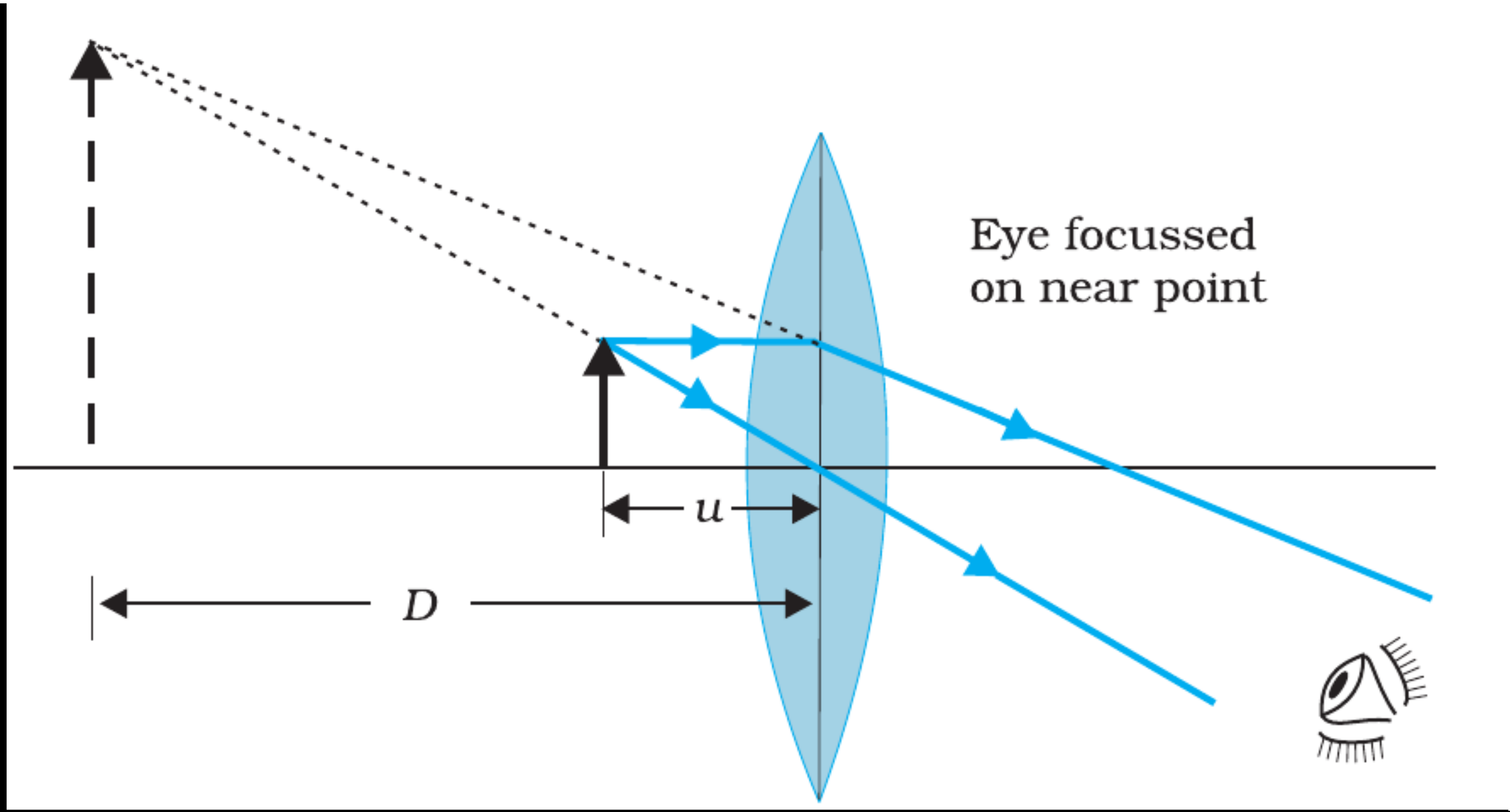
$$m = \frac{\theta_i}{\theta_0} = \frac{\frac{\cancel{h}}{f}}{\frac{\cancel{h}}{D}}$$

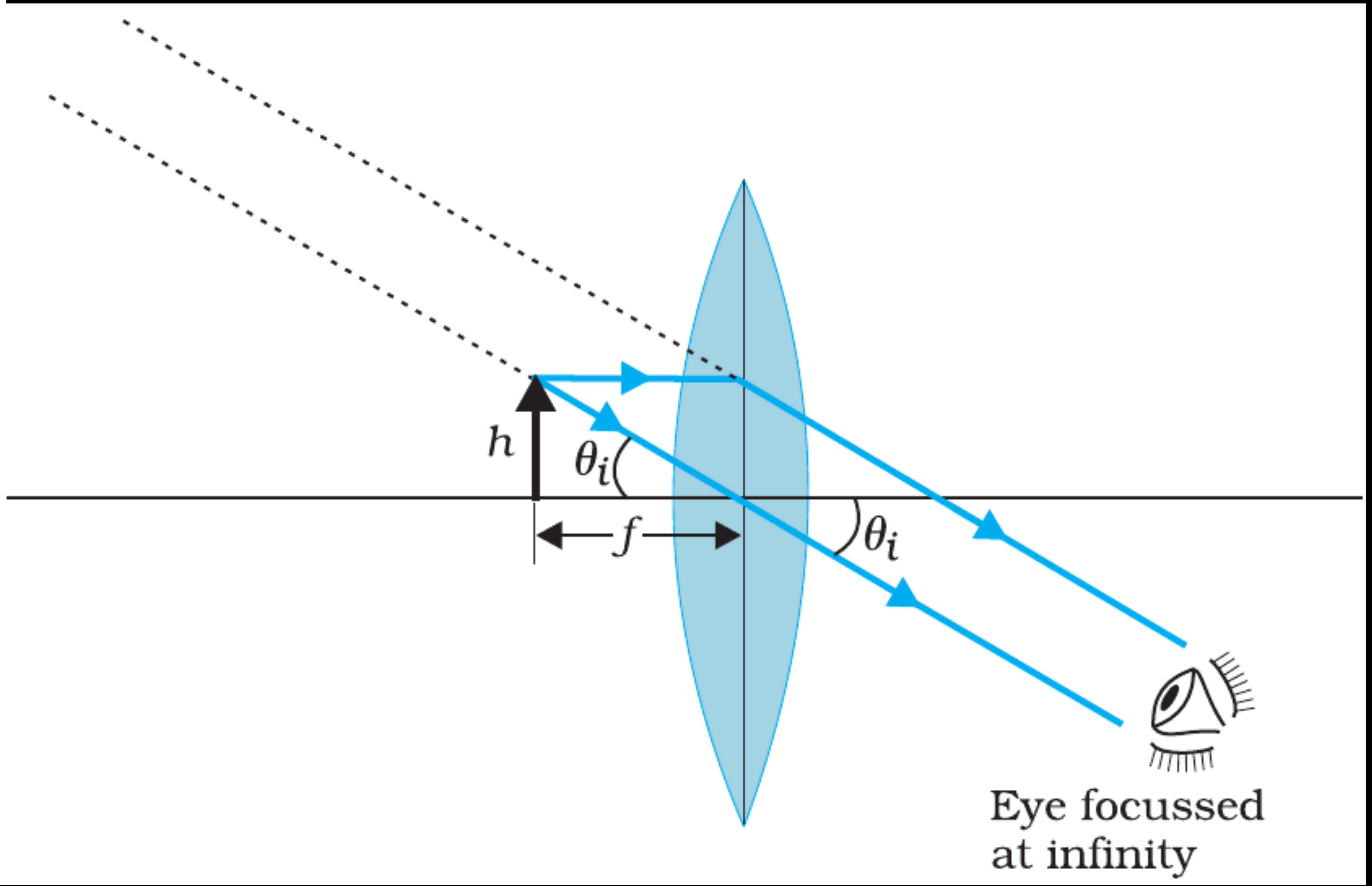
$$\boxed{m = \frac{D}{f}}$$

→ 1 less than when image is formed at near point.
But, veiwing is more comfortable.

Simple Microscope







Compound Microscope

* A simple microscope has a limited maximum magnification (≤ 9) for realistic focal lengths.

* For much larger magnifications, one uses two lenses, one compounding the effect of the other. This is known as a compound microscope.

* The lens nearest the object, called the objective, forms a real, inverted, magnified image of the object.

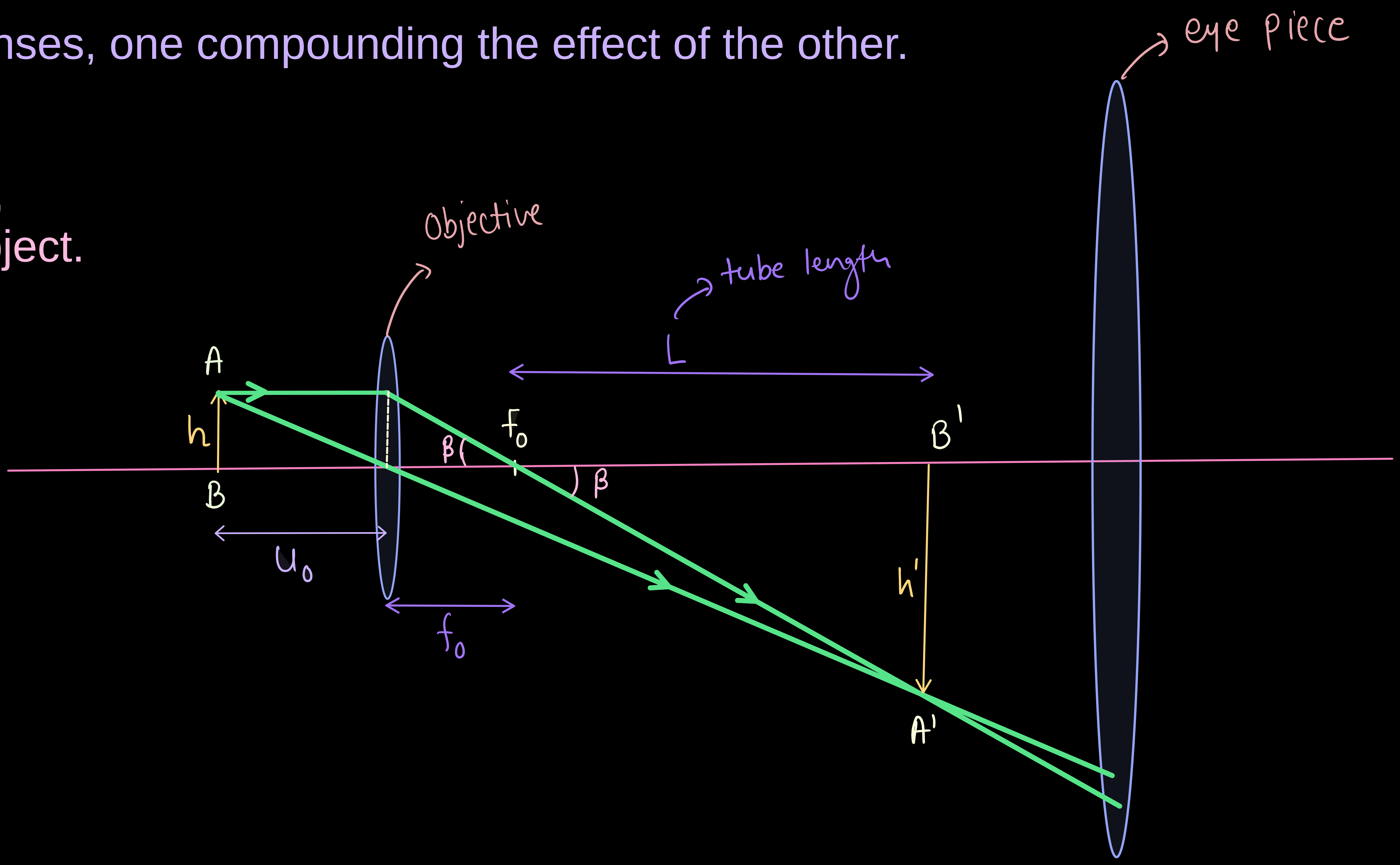
* Linear magnification (m_o) due to objective lens-

$$m_o = \frac{-h'}{h}$$

$$\tan\beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$\boxed{m_o = -\frac{L}{f_o}} \quad \text{— (1)}$$



Compound Microscope

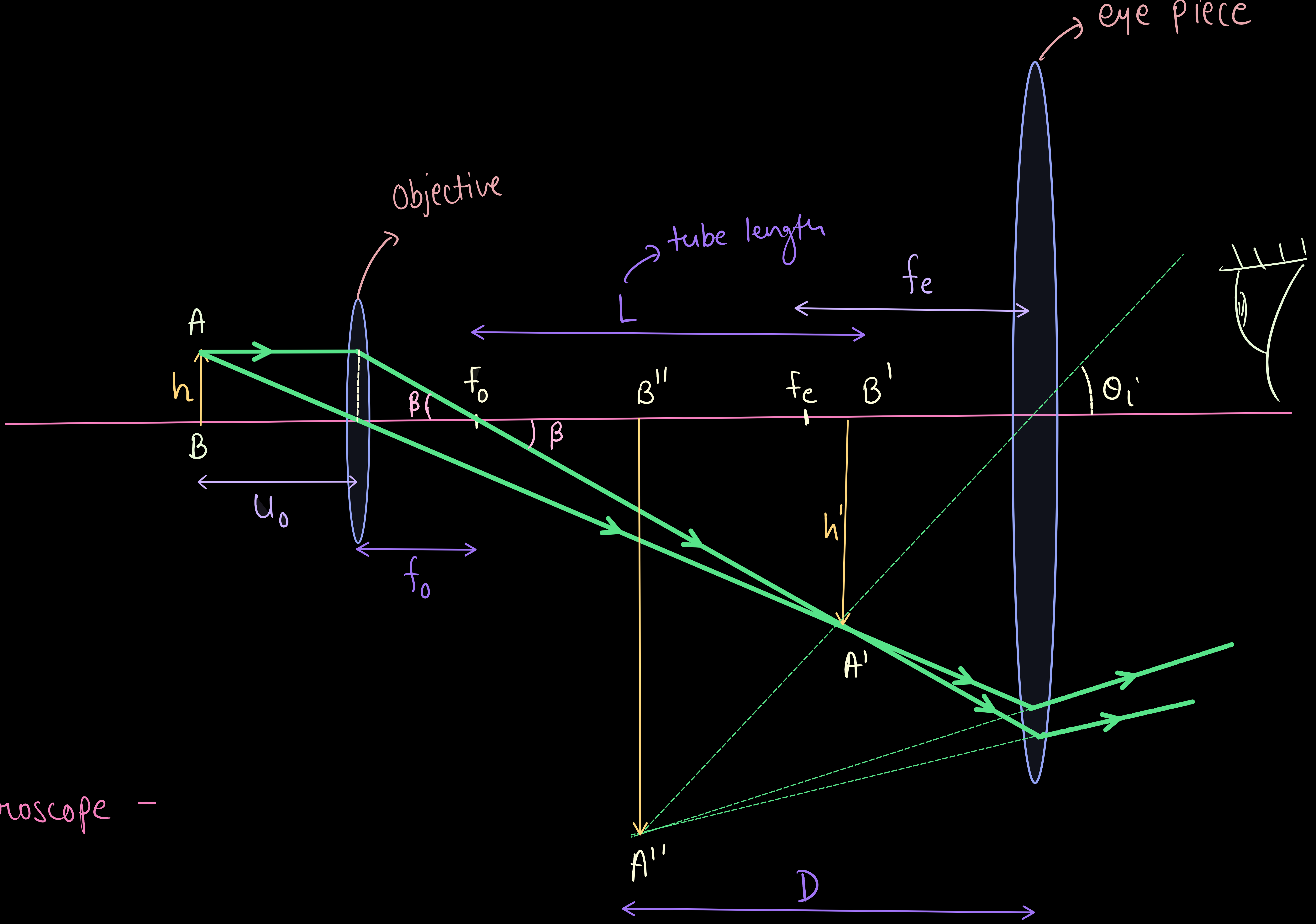
* Linear magnification (m_o) due to objective lens-

$$m_o = \frac{-h'}{h}$$

$$\tan\beta = \frac{h}{f_o} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_o}$$

$$\boxed{m_o = -\frac{L}{f_o}} \quad \text{--- (1)}$$



* Case I - When image is formed at near point (LDDV)

magnification by eye-piece -

$$m_e = 1 + \frac{D}{f_e} \quad \text{--- (2)}$$

⇒ Total Magnification produced by compound microscope -

$$m = m_o \times m_e$$

$$\boxed{m = -\frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)} \quad \text{--- (3)} \quad ★★$$

Compound Microscope

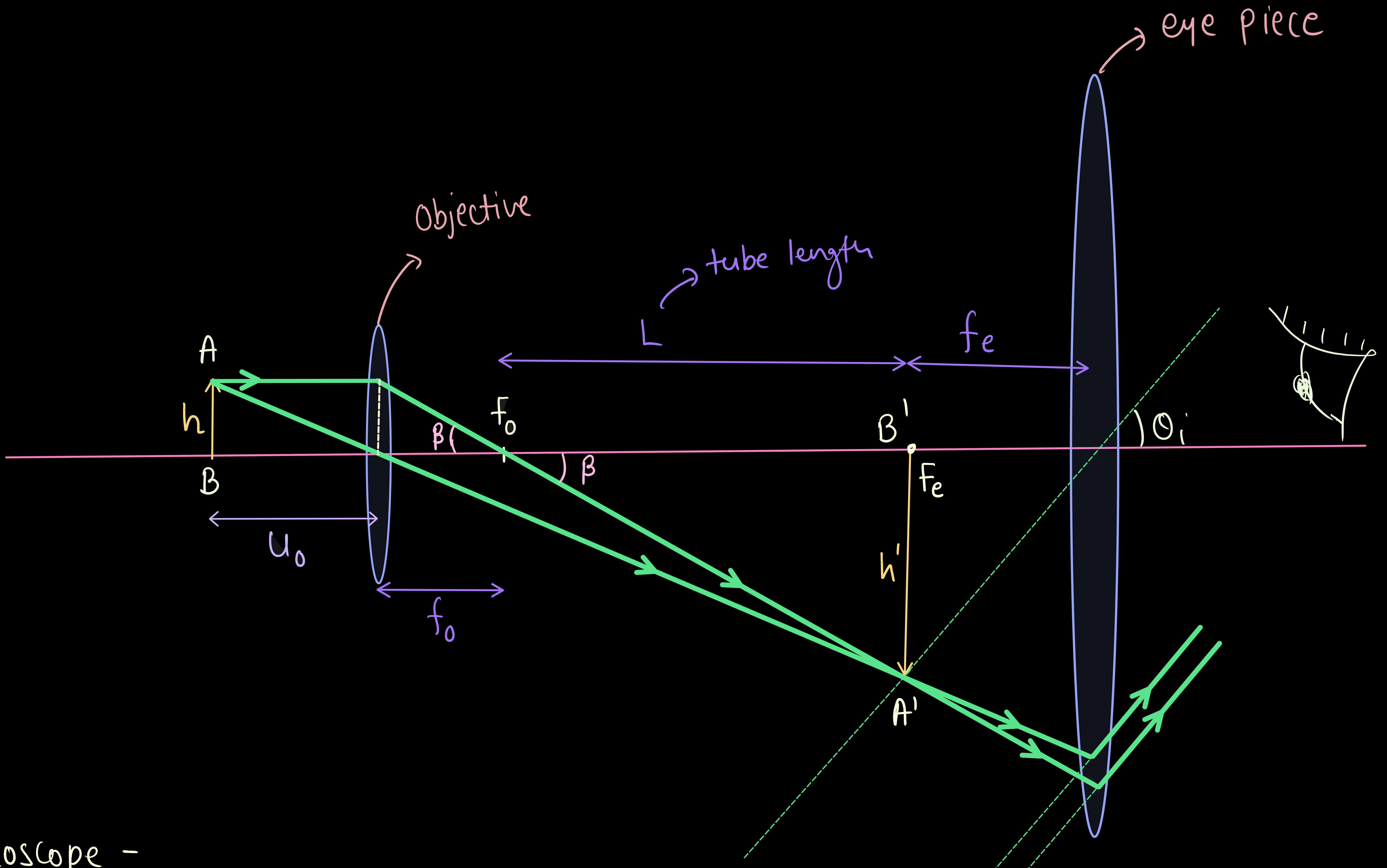
* Linear magnification (m_0) due to objective lens-

$$m_0 = \frac{-h'}{h}$$

$$\tan\beta = \frac{h}{f_0} = \frac{h'}{L}$$

$$\frac{h'}{h} = \frac{L}{f_0}$$

$$\boxed{m_0 = \frac{-L}{f_0}} \quad \text{— (1)}$$



Case II- when image is formed at infinity-

magnification by eyepeice lens.

$$m_e = \frac{D}{f_e} \quad \text{— (2)}$$

Total magnification produced by compound microscope -

$$m = m_0 \times m_e$$

$$\boxed{m = \frac{-L}{f_0} \times \frac{D}{f_e}} \quad \text{— (3)} \quad ☆☆$$

* Clearly, to achieve a large magnification of a small object (hence the name microscope), the objective and eyepiece should have small focal lengths. In practice, it is difficult to make the focal length much smaller than 1 cm. Also large lenses are required to make L large.

Astronomical Telescope

* The telescope is used to provide angular magnification of distant objects.

* It also has an objective and an eyepiece.

* But here, the objective has a large focal length and a much larger aperture than the eyepiece.

magnifying power $m = \dfrac{\beta}{\alpha}$

$$m = \frac{\frac{\cancel{h}}{f_e}}{\frac{\cancel{h}}{f_o}}$$

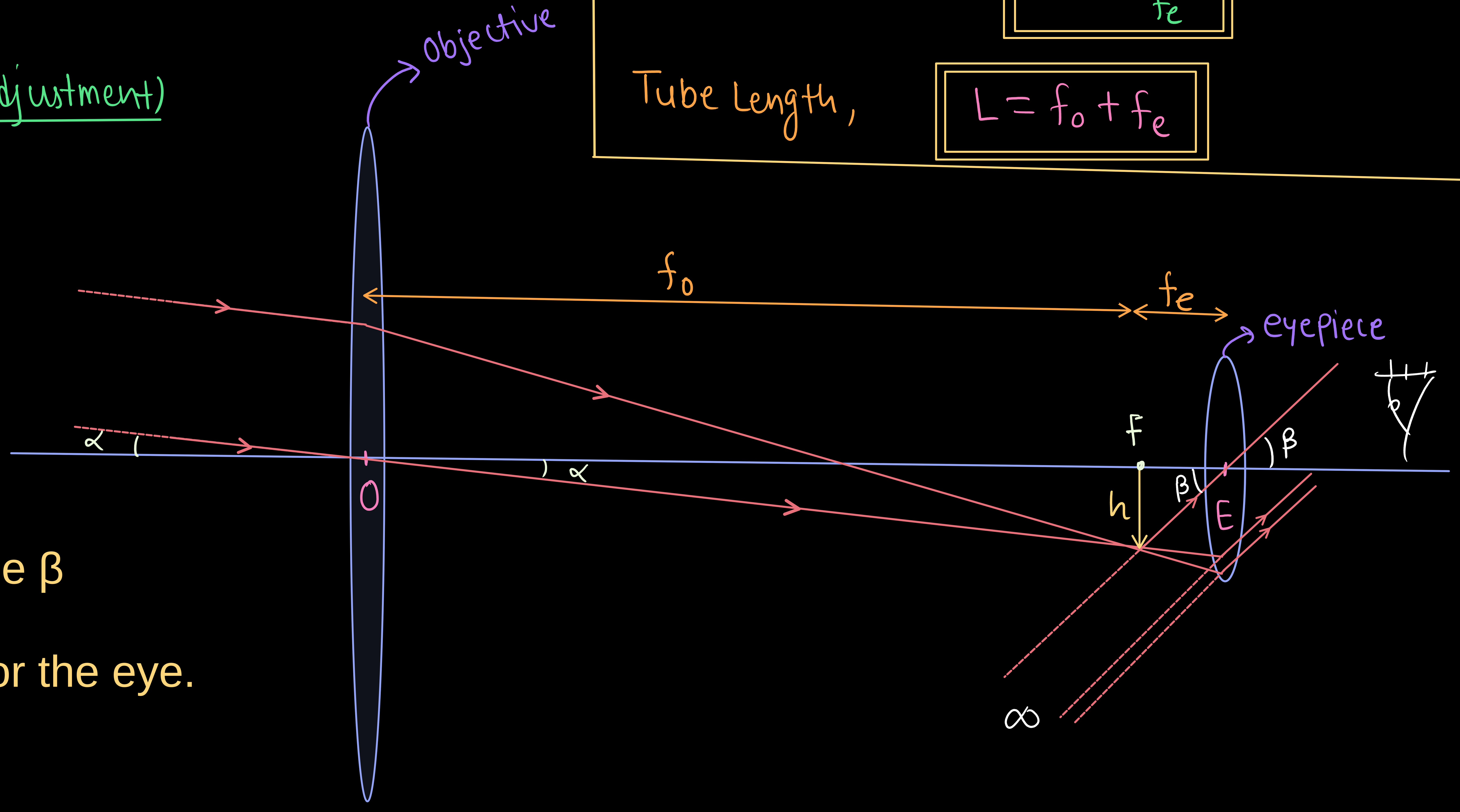
$$\boxed{m = \frac{f_o}{f_e}}$$

Tube Length, $\boxed{L = f_o + f_e}$

Case I – When image is formed at infinity (Normal Adjustment)

* Light from a distant object enters the objective and a real image is formed in the tube at its second focal point.

* The eyepiece magnifies this image producing a final inverted image at ∞.



* The magnifying power m is the ratio of the angle β subtended at the eye by the final image to the angle α which the object subtends at the lens or the eye.

$$\tan\beta \approx \beta = \frac{h}{f_e}, \quad \tan\alpha \approx \alpha = \frac{h}{f_o}$$

Astronomical Telescope

* The main considerations with an astronomical telescope are its light gathering power and its resolution or resolving power. The former clearly depends on the area of the objective. With larger diameters, fainter objects can be observed. The resolving power, or the ability to observe two objects distinctly, which are in very nearly the same direction, also depends on the diameter of the objective. So, the desirable aim in optical telescopes is to make them with objective of large diameter.

* Terrestrial telescopes have, in addition, a pair of inverting lenses to make the final image erect.

* Problems with refracting telescopes -

Big lenses of refracting telescopes tend to be very heavy and therefore, difficult to make and support by their edges.

Further, it is rather difficult and expensive to make such large sized lenses which form images that are free from any kind of chromatic aberration and distortions.

Reflecting Telescope

* For these reasons, modern telescopes use a concave mirror rather than a lens for the objective.

* Telescopes with mirror objectives are called reflecting telescopes.

* There is no chromatic aberration in a mirror.

* Mechanical support is much less of a problem since a mirror weighs much less than a lens of equivalent optical quality, and can be supported over its entire back surface, not just over its rim.

* One obvious problem with a reflecting telescope is that the objective mirror focusses light inside the telescope tube.

* One must have an eyepiece and the observer right there, obstructing some light (depending on the size of the observer cage).

* Solution to the problem is to deflect the light being focussed by convex secondary mirror to focus the incident light, which now passes through a hole in the objective primary mirror. This is known as a Cassegrain telescope.

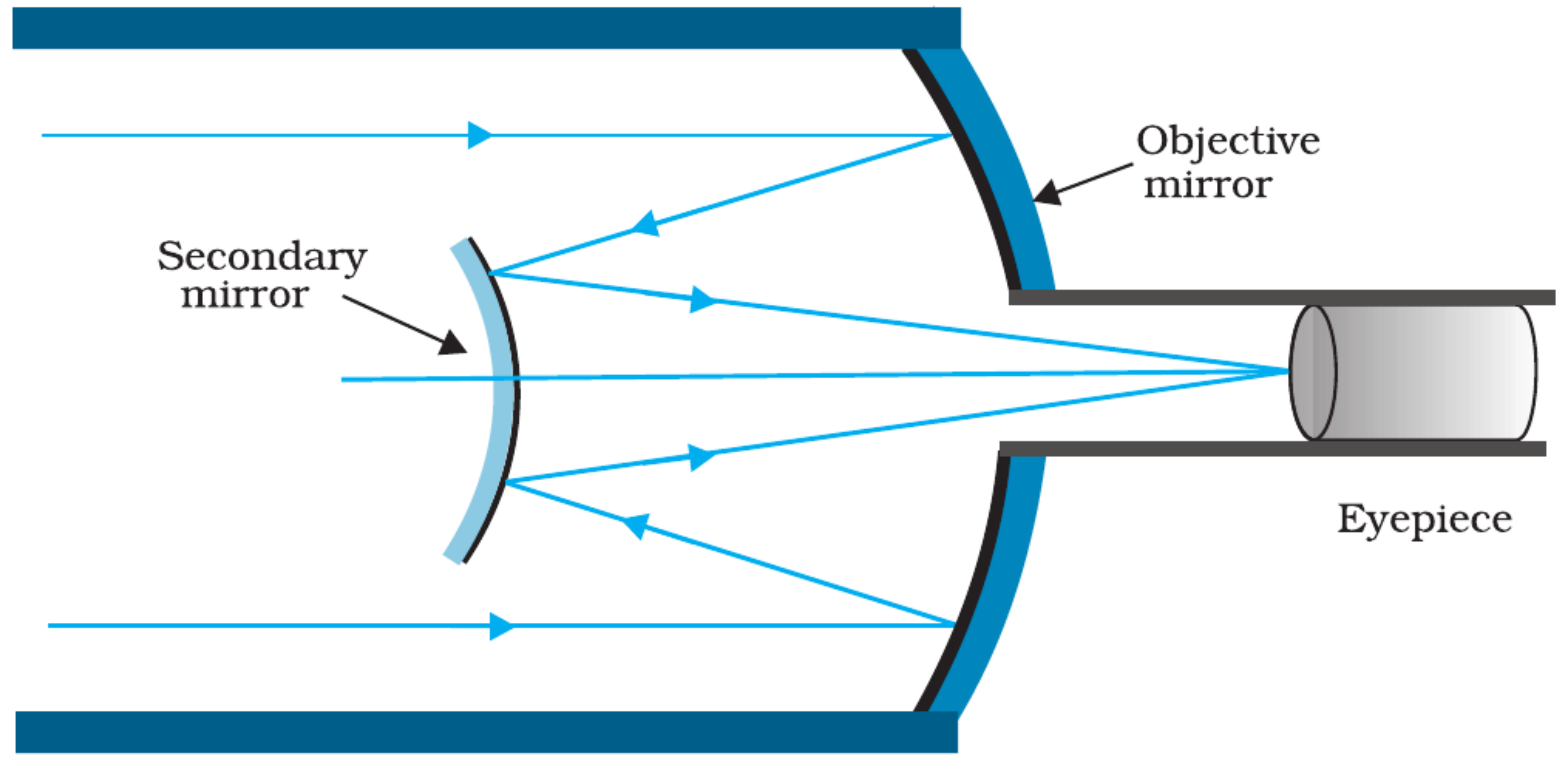


FIGURE 9.30 Schematic diagram of a reflecting telescope (Cassegrain).